



VIRGINIA COMMONWEALTH UNIVERSITY

Statistical Analysis and Modelling (SCMA 632)

A7a: Volatility Modelling and Multivariate Time Series Analysis

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Introduction

In this report, we undertake a comprehensive analysis of financial market volatility and commodity price dynamics using advanced time series econometric methods. The study is divided into two main parts.

The first part focuses on modelling the volatility of NVIDIA Corporation's (NVDA) daily stock returns over a five-year period. Employing ARCH and GARCH models, we capture the characteristic volatility clustering observed in financial returns. Model estimation, diagnostic testing, and comparison using information criteria guide the selection of the appropriate volatility model. Additionally, we provide a forecast of future volatility over a 90-day horizon to assess expected market risk.

The second part investigates the interrelationships among major global commodity prices using multivariate time series models. Utilizing monthly data from the World Bank Pink Sheet, we classify commodities into distinct categories—Energy, Metals, and Food—and conduct Vector Autoregression (VAR) and Vector Error Correction Model (VECM) analyses. These allow us to capture both short-run dynamics and long-run equilibrium relationships via stationarity testing, lag order selection, cointegration testing, and impulse response analysis.

Objectives

The primary objectives of this report are to:

- **Model and forecast volatility** of NVIDIA Corporation (NVDA) stock returns using ARCH and GARCH frameworks, capturing the temporal clustering of volatility commonly observed in financial markets.
- **Conduct a multivariate time series analysis** of global commodity prices, grouped into Energy, Metals, and Food categories, to uncover the dynamic interrelationships and long-run equilibrium conditions among commodity price indices.
- **Apply rigorous statistical testing** including stationarity tests, lag order selection, and Johansen cointegration testing to determine appropriate model specifications within each commodity group.
- **Estimate and interpret VAR and VECM models** to analyze both short-term price interactions and long-term cointegrating relationships in commodity markets.
- **Generate impulse response functions and forecasts** to illustrate the transmission of shocks within commodity categories and assess future volatility in stock returns.
- **Demonstrate comprehensive data preprocessing and cleaning techniques** to handle real-world data challenges such as missing values and non-numeric entries, ensuring robustness and reproducibility of the analyses.

Business Significance

Volatility Modelling (Part A)

Understanding and forecasting volatility is central to effective financial risk management, trading, and investment decisions. By modelling the volatility of NVIDIA Corporation (NVDA) stock returns with ARCH and GARCH frameworks, this analysis provides:

1. **Quantitative risk assessment:** The estimated conditional volatility offers a data-driven foundation for measuring potential price swings, which is essential for setting appropriate risk limits, pricing derivatives, and determining hedging strategies.
2. **Strategic decision-making:** Volatility forecasts—particularly over horizons such as 90 days—equip portfolio managers and traders with actionable insights to adjust their asset allocations or risk exposures in response to anticipated periods of higher or lower uncertainty.
3. **Evaluation of market stability:** Persistent volatility clustering, as revealed in the GARCH model, signals underlying market conditions and helps investors anticipate shifts that could impact portfolio returns.
4. **Informing stakeholders:** Transparent volatility estimates support more reliable communications with stakeholders regarding expected risk and potential drawdowns.

Multivariate Commodity Price Analysis (Part B)

Analyzing commodity prices using multivariate time series models (VAR and VECM) delivers significant business value for stakeholders in international trade, supply chain management, policymaking, and investment:

1. **Market interdependency mapping:** By quantifying short-term and long-term relationships among commodities within sectors (Energy, Metals, Food), businesses and policymakers can:
 - a. Identify potential transmission pathways for shocks—such as how a price change in crude oil may influence natural gas or coal prices.
 - b. Understand which commodities act as price leaders or followers, informing procurement and sales strategies.
2. **Risk diversification and hedging:** Cointegration analysis reveals whether commodities tend to move together over the long run; this knowledge enhances the design of hedging strategies and diversification of procurement/inventory.
3. **Scenario analysis and forecasting:** VAR and VECM models enable firms to simulate and anticipate the effect of global events (e.g., geopolitical tensions, policy shifts, supply disruptions) on interconnected commodity markets.
4. **Informed pricing and budgeting:** Firms and public bodies can leverage forecasted relationships to improve pricing strategies, negotiate contracts, and make robust budget projections based on expected price co-movements and volatility.
5. **Policy insights:** Regulators and policymakers gain evidence of structural linkages which can justify interventions (such as stockpiling or releasing reserves) or anticipate spillover effects from shocks in one commodity to others.

Overall Impact

1. **Actionable intelligence:** The rigorous approach adopted ensures that stakeholders are not only reactive but can be proactive in managing risks and capitalizing on market opportunities.
2. **Credibility and robustness:** Empirical evidence produced through advanced econometric modelling increases confidence in business decisions and policy recommendations.
3. **Template for future analyses:** The methodologies demonstrated can be adapted to other assets, time periods, or markets, providing a scalable framework for ongoing business analytics and strategy development.

Results

Part A: Volatility Modelling of NVDA Stock Returns

- The ARCH LM test conducted on NVDA daily log returns indicated marginal evidence of conditional heteroskedasticity, supporting the application of ARCH/GARCH models.
- Both ARCH(1) and GARCH(1,1) models were estimated. The GARCH(1,1) model demonstrated better fit, having a lower Akaike Information Criterion (AIC) and statistically significant parameters.
- The GARCH(1,1) model estimated notable persistence in volatility, with the sum of ARCH and GARCH coefficients approaching one.
- A 90-day ahead forecast of conditional volatility was generated, showing a volatility path consistent with recent historical levels.

Part B: Multivariate Time Series Analysis of Global Commodity Prices

- Monthly commodity price data were pre-processed, resulting in cleansed datasets with numeric values and no missing entries per commodity group.
- Augmented Dickey-Fuller tests on differenced log prices confirmed stationarity across all commodity series analyzed.
- Lag order selection via the Akaike Information Criterion identified optimal VAR lag lengths mostly ranging between 2 and 6 months, varying by commodity category.
- VAR models were successfully fitted for each commodity group and subgroup, with model summaries indicating significant interdependencies among prices within groups such as Crude Oil, Natural Gas, Coal, Metals, Wheat, and Sugar.
- Impulse response function plots were generated for each VAR model, illustrating the dynamic response of commodity prices to shocks within their categories over a 10-month horizon.
- Johansen cointegration tests detected statistically significant cointegration ranks primarily in the Crude Oil and Metals groups.

- VECM models were estimated for these groups, with model summaries indicating significant adjustment coefficients confirming long-run equilibrium relations among prices.

Interpretations

Part A: Volatility Modelling of NVDA Stock Returns

The marginal evidence of ARCH effects in NVIDIA stock returns highlights the presence of volatility clustering—a common trait in financial markets where periods of high volatility tend to be followed by similar periods. The superiority of the GARCH(1,1) model over the simpler ARCH(1) reflects the importance of capturing both immediate shocks (ARCH effects) and persistent volatility components (GARCH effects) to accurately model return variability. The near-unity sum of the ARCH and GARCH parameters suggests strong persistence in volatility, indicating that shocks to returns have prolonged effects on future volatility.

The 90-day ahead forecast reveals that the expected volatility of NVDA remains relatively stable, suggesting no immediate spikes in uncertainty or market turbulence. This forecast can inform risk management strategies by providing anticipated measures of market risk over the short to medium term.

Part B: Multivariate Time Series Analysis of Global Commodity Prices

The confirmation of stationarity after differencing logged commodity prices validates the use of VAR and VECM frameworks, which require stationary input series. Optimal lag lengths between 2 and 6 months show that commodity price dynamics capture influences from several preceding months, consistent with market adjustments and transmission delays.

VAR model outcomes reveal significant short-run interdependencies among commodity prices within each group. For instance, in the Crude Oil subgroup, price shocks in one benchmark (e.g., Dubai crude) have appreciable short-term impacts on others such as Brent and WTI. Similarly, metals and agricultural commodities show intricate webs of influence, underscoring complex market relationships driven by factors such as substitution effects, regional supply-demand conditions, and transportation costs.

The presence of cointegration in the Crude Oil and Metals categories points to stable, long-run equilibrium relationships in these markets. This implies that despite short-term fluctuations, prices in these groups tend to move together over time, reflecting fundamental linkages such as shared production costs, cross-market arbitrage, or global demand patterns. The significant adjustment coefficients in the VECM models suggest that deviations from equilibrium are corrected efficiently, indicating market mechanisms that restore balance aftershocks.

In contrast, the absence or lower ranks of cointegration in other commodity categories suggest less stable long-term relationships, possibly due to greater regional heterogeneity or exposure to idiosyncratic shocks. These markets may be more affected by short-term supply disruptions, policy interventions, or localized demand changes.

Overall, the multivariate framework captures both the immediate transmission of shocks within commodity groups and the long-term co-movements, providing a nuanced understanding of commodity price dynamics essential for stakeholders in trading, policy, and supply chain management.

Recommendations

Part A: Volatility Modelling of NVDA Stock

- 1. Incorporate Diagnostics for Model Validation:**
Regularly perform residual diagnostics on fitted ARCH/GARCH models, including testing for remaining ARCH effects, normality, and autocorrelation in standardized residuals to ensure model adequacy.
- 2. Explore Alternative Volatility Models:**
Consider extending the analysis to more flexible volatility models such as EGARCH or GJR-GARCH to capture asymmetries and leverage effects often observed in equity returns.
- 3. Use Forecasts in Risk Management:**
Integrate volatility forecasts into portfolio risk assessment frameworks, such as Value-at-Risk (VaR) calculations, to enhance risk monitoring and capital allocation.
- 4. Update Model Periodically:**
As market conditions evolve, routinely update volatility models with recent data to maintain forecast accuracy and adapt to regime changes.

Part B: Multivariate Commodity Price Analysis

- 1. Implement Comprehensive Model Diagnostics:**
Enhance analysis robustness by performing serial correlation tests (e.g., Portmanteau test), normality tests (e.g., Jarque–Bera), and stability checks for VAR/VECM residuals, ensuring model validity before forecasting or policy simulation.
- 2. Incorporate Forecasting and Scenario Analysis:**
Extend the VAR/VECM framework to generate multi-step ahead forecasts and conduct scenario analyses of shocks to commodity prices, aiding stakeholders in strategic planning and risk mitigation.
- 3. Consider Structural VAR (SVAR) Extensions:**
To identify causal relationships, structural breaks, or contemporaneous interactions, apply SVAR or regime-switching models for richer economic interpretation.
- 4. Address Data Limitations:**
Explore imputation or interpolation techniques for missing data instead of listwise deletion to preserve sample size and improve model estimates.
- 5. Expand Commodity Coverage and Granularity:**
Include additional relevant commodities or disaggregate by geography to capture localized effects and improve relevance for specific market participants.

6. **Integrate Macroeconomic Variables:**

Augment the commodity models with relevant macroeconomic indicators (e.g., exchange rates, inflation rates, industrial production) to understand broader economic drivers.

7. **Regular Re-estimation:**

Given changing market dynamics, periodically re-estimate VAR/VECM models to incorporate new data and maintain model accuracy.

References

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- World Bank. (2025). *Commodity Markets Outlook: The Pink Sheet*. Retrieved from <https://thedocs.worldbank.org/en/doc/18675f1d1639c7a34d463f59263ba0a2-0050012025/related/CMO-Historical-Data-Monthly.xlsx>